Name

Mathematics Stage 5 Paths

Factorising Negative and Algebraic Factors

Book 1

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Syllabus Content

MA5-ALG-P-01 simplifies algebraic fractions involving indices, and expands and factorises algebraic expressions

Factorise algebraic expressions by taking out a common algebraic factor

- Factorise algebraic expressions, including those involving indices, by determining the highest common factors (HCF) including negative coefficients/pronumerals

Factorising a Negative HCF

Factorising a Negative HCF

To **factorise** with a **negative HCF**:

1. Identify the negative **highest common factor** (HCF) of the terms.
   Write it outside brackets.
2. Divide each term by the HCF. Put the results inside the brackets.

You should factor out a negative HCF if both terms are negative, or if the question asks for it.
This skill is used in a later topic (factorising by grouping in pairs).

Example Factorise by finding a finding a negative HCF

Factorise these expressions by finding a negative HCF.

$-8x - 16$

1. $-8(\quad)$
2. $-8(x + 2)$

$12x - 4xy$

1. $-4x(\quad)$
2. $-4x(-3 + y)$

Guided Practice

Factorise these expressions by finding a negative HCF.

a $-8x - 4$

$= -4(\quad)$

$-4(2x + 1)$

b $-2m + 4n$

$= -2(\quad)$

$-2(m - 2n)$

c $-3xy + 18x$

$-3x(y - 6)$

d $8x^2 - 12x$

$-4x(-2x + 3)$

e $6x - 20x^2$

$-2x(-3 + 10x)$

f $5a^3x + 10a^2$

$-5a^2(-ax - 2)$

1 Complete these factorisations.

a $7x + 7 = 7(___ + ___)$

$x, 1$

b $14x + 7 = 7(___ + ___)$

$2x, 1$

c $14x - 7 = 7(___ - ___)$

$2x, 1$

d $-9x + 9 = -9(___ - ___)$

$x, 1$

e $-9x + 9 = 9(___ + ___)$

$-x, 1$

f $6 - 12x = 6(___ - ___)$

$1, 2x$

2 Factorise by taking out a negative HCF.

a $-8x - 4$

$-4(2x + 1)$

b $-4x - 2$

$-2(2x + 1)$

c $-10x - 5y$

$-5(2x + y)$

d $-7a - 14b$

$-7(a + 2y)$

e $-9x - 1$

$-(9x + 1)$

f $-6y - 8$

$-2(3y + 4)$

g $-10x - 15y$

$-5(2x + 3y)$

h $-4m - 20n$

$-4(m + 5n)$

i $-3x^2 - 18x$

$-3x(x + 6)$

j $-6x + 8$

$-2(3y - 4)$

k $-12x + 16$

$-4(3x - 4)$

l $-60x + 45$

$-15(4x + 3)$

m $-9x - 12$

$-3(3x + 4)$

n $-20xy + 15x$

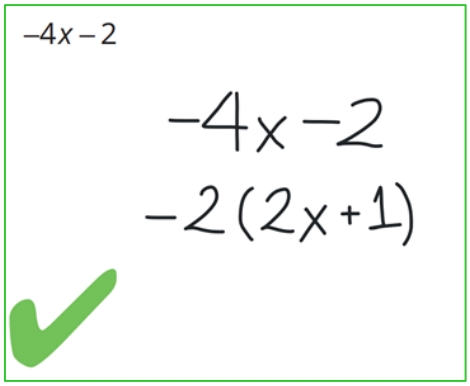
$-5x(4y - 3)$

o $21x - 15xy$

$3x(7 - 5y)$

3 Read the examples carefully and answer the questions.

Fei's work



Where did the +1 come from in Fei's factored expression?

-2 divided by …… is 1

Factorise:

$-9x^2 - 3x$

$-3x(3x + 1)$

4 Factorise the following. For these questions, always take out a negative HCF.

a	$3x - 18$	**b**	$-12a - 3$	**c**	$-5x - 30$
	$-3(-x + 6)$		$-3(4a + 1)$		$-5(x + 6)$
d	$10a - 5a^2$	**e**	$4x^2 + x$	**f**	$6b^2 - 18b$
	$-5a(-2 + a)$		$-x(-4x - 1)$		$-6b(-b + 3)$
g	$12x - 30x^2$	**h**	$-2x - x^2$	**i**	$-4y - 8y^2$
	$-6x(-2 + 5x)$		$-x(2 + x)$		$-4y(1 + 2y)$
j	$6xyz^2 + 3z^2$	**k**	$2x^2yz - 4xy$	**l**	$-12m^2n - 12mn^2$
	$-3z^2(-2xy - 1)$		$-2xy(-xz + 2)$		$-12mn(m + n)$
m	$-16y^2 - 6y$	**n**	$-5a^2 - 10a$	**o**	$-6x - 20x^2$
	$-2y(8y + 3)$		$-5a(a + 2)$		$-2x(3 + 10x)$
p	$-6p - 15p^2$	**q**	$-16b - 8b^2$	**r**	$-9x - 27x^2$
	$-3p(2 + 5p)$		$-8b(2 + b)$		$-9x(1 + 3x)$

5 Many of the following expressions are not fully factorised. Finish the factorisation.

a	$2(3x + 9)$ $= 2 \times 3(x + 3)$ $=$ $6(x + 3)$	**f**	$2(6x - 12)$ $12(x - 2)$	**k**	$4(12 - 6x + 4y)$ $8(6 - 3x + 2y)$
b	$2(-3x - 9)$ $-6(x + 3)$	**g**	$2(12 - 6x)$ $12(2 - x)$	**l**	$4(12 - 6x + 5y)$ can't factorise further
c	$2(6x + 10)$ $4(3x + 5)$	**h**	$4(-12 - 6x)$ $-24(2 + x)$	**m**	$4(12x - 6x^2 + 5xy)$ $4x(12 - 6x + 5y)$
d	$2(6x + 11)$ can't factorise further	**i**	$4(12 - 6x + 2y)$ $8(6 - 3x + y)$	**n**	$4(12x - 6x^2 + 4xy)$ $8x(6 - x^2 + 2y)$
e	$2(-6x - 12)$ $-12(x + 2)$	**j**	$4(12 - 6x + 3y)$ $12(4 - 2x + y)$	**o**	$4(12x^2 - 6x^2 + 5x^2y)$ $4x^2(6 + 5y)$

6 Common factors can also be removed from expressions with more than two terms.
For example: $2x^2 + 6x + 10xy = 2x(x + 3 + 5y)$. Factorise these expressions.

a	$3a + 9b + 12$	**b**	$5z^2 - 10z + zy$	**c**	$x^2 - 2xy + x^2y$
	$3(a + 3b + 4)$		$z(5z - 10 + y)$		$x(x - 2y + xy)$
d	$4by - 2b + 6b^2$	**e**	$-12xy - 8yz - 20xyz$	**f**	$3ab + 4ab^2 + 6a^2b$
	$2b(2y - 1 + 3b)$		$-4y(3x + 2z + 5xz)$		$ab(3 + 4b + 6a)$

7 Sometimes we can choose to factor out a negative or a positive HCF.
Both factorisations are correct. For example:

$$-13x + 26 = -13(x - 2) \quad \text{(HCF is } -13\text{)}$$
$$\text{or} \quad -13x + 26 = 13(-x + 2) \quad \text{(HCF is } 13\text{)}$$
$$= 13(2 - x)$$

Factorise these expressions in two ways.

a	$-4x + 12$	**b**	$-3x + 9$	**c**	$-8n + 8$
	$4(-x + 3)$, $-4(x - 3)$		$3(-x + 3)$, $-3(x - 3)$		$8(-n + 1)$, $-8(n - 1)$
d	$-5m + 5m^2$	**e**	$-7x + 7x^2$	**f**	$-4y + 22y^2$
	$5m(-1 + m)$, $-5m(1 - m)$		$7x(-1 + x)$, $-7x(1 - x)$		$2y(-2 + 11y)$, $-2y(2 - 11y)$
g	$-8n + 12n^2$	**h**	$-8y + 20$	**i**	$-15mn + 10$
	$4n(-2 + 3n)$, $-4n(2 - 3n)$		$4(-2y + 5)$, $-4(2y - 5)$		$5xy(-2y + 3x)$, $-5xy(2y - 3x)$
j	$9x - 27x^2$	**k**	$16ab - 8ab^2$	**l**	$-10xy^2 + 15x^2y$
	$9x(1 - 3x)$, $-9x(-1 + 3x)$		$8ab(2 - b)$, $-8ab(-2 + b)$		$5xy(-2y + 3x)$, $-5xy(2y - 3x)$
m	$x^2y - 4xy + xy^2$	**n**	$6ab - 10a^2b + 8ab^2$	**o**	$-3a^3b + 12ab^2 + 9a^2b^3$
	$xy(x - 4 + y)$, $-xy(-x + 4 - y)$		$2ab(3 - 5a + 4b)$, $-2ab(-3 + 5a - 4b)$		$3ab(-a^2 + 4b + 3ab^2)$, $-3ab(a^2 - 4b - 3ab^2)$